Proof. (i) Let Y be an algebraic representation of $GL(V)$. We have an embedding $\xi : Y \to Y \otimes R$ given by $(u, \xi(v))(g) := u(gv)$, $u \in Y^*$. It is easy to see that ξ is a homomorphism of representations (where the action of $GL(V)$ on the first component of $Y \otimes R$ is trivial). Thus, it suffices to prove the theorem for a subrepresentation $Y \subset R^m$. Now, every element of R is a polynomial of g_{ij} times a nonpositive power of $\det(g)$. Thus, R is a quotient of a direct sum of representations of the form $S^r(V \otimes V^*) \otimes (\wedge^N V^*)^{\otimes s}$, where the group action on V^* in the product $V \otimes V^*$ is trivial. So we may assume that Y is contained in a quotient of a (finite) direct sum of such representations. Thus, Y is contained in a direct sum of representations of the form $V^{\otimes n} \otimes (\wedge^N V^*)^{\otimes s}$, and we are done.

(ii) Let Y be an algebraic representation of $GL(V)$, and let us regard R as a representation of $GL(V)$ via $(\rho(h)\phi)(x) = \phi(xh)$. Then $\mathrm{Hom}_{GL(V)}(Y, R)$ is the space of polynomial functions f on $GL(V)$ with values in Y^* which are right $GL(V)$-equivariant (i.e., such that $f(xg) = g^{-1}f(x)$). This space is naturally identified with Y^*. Taking into account the proof of (i), we deduce that R has the required decomposition, which is compatible with the second action of $GL(V)$ (by left multiplications). This implies the statement. $\square$

Remark 5.23.3. Since the Lie algebra $\mathfrak{sl}(V)$ of traceless operators on V is a quotient of $\mathfrak{gl}(V)$ by scalars, the above results extend in a straightforward manner to representations of the Lie algebra $\mathfrak{sl}(V)$. Similarly, the results for $GL(V)$ extend to the case of the group $SL(V)$ of operators with determinant 1. The only difference is that in this case the representations L_λ and $L_{\lambda+1^m}$ are isomorphic, so the irreducible representations are parametrized by integer sequences $\lambda_1 \geq \cdots \geq \lambda_N$ up to a simultaneous shift by a constant.

In particular, one can show that any finite dimensional representation of $\mathfrak{sl}(V)$ is completely reducible and any irreducible representation is of the form L_λ (we will not do this here). For $\dim V = 2$ one then recovers the representation theory of $\mathfrak{sl}(2)$ studied in Problem 2.15.1.